

A Technical Introduction to Bayesian network

$$P(h | D) = \frac{P(D | h)P(h)}{P(D)}$$

Section 1

Graphical model Formally, Bayesian networks are directed acyclic graphs (DAGs) whose nodes represent variables in the Bayesian sense: they may be observable quantities, latent variables, unknown parameters or hypotheses. Each edge represents a direct conditional dependency. Any pair of nodes that are not connected (i.e. no path connects one node to the other) represent variables that are conditionally independent of each other. Each node is associated with a probability function that takes, as input, a particular set of values for the node's parent variables, and gives (as output) the probability (or probability distribution, if applicable) of the variable represented by the node. For example, if m parent nodes represent m Boolean variables, then the probability function could be represented by a table of 2^m entries, one entry for each of the 2^m possible parent combinations. Similar ideas may be applied to undirected, and possibly cyclic, graphs such as Markov networks.

Section 2

A back-door path is one that ends with an arrow into X . Sets that satisfy the back-door criterion are called "sufficient" or "admissible." For example, the set $Z = R$ is admissible for predicting the effect of $S = T$ on G , because R d-separates the (only) back-door path $S \leftarrow R \rightarrow G$. However, if S is not observed, no other set d-separates this path and the effect of turning the sprinkler on ($S = T$) on the grass (G) cannot be predicted from passive observations. In that case $P(G | \text{do}(S = T))$ is not "identified". This reflects the fact that, lacking interventional data, the observed dependence between S and G is due to a causal connection or is spurious (apparent dependence arising from a common cause, R). (see Simpson's paradox) To determine whether a causal relation is identified from an arbitrary Bayesian network with unobserved variables, one can use the three rules of "do-calculus" and test whether all do terms can be removed from the expression of that relation, thus confirming that the desired quantity is estimable from frequency data. Using a Bayesian network can save considerable amounts of memory over exhaustive probability tables, if the dependencies in the joint distribution are sparse. For example, a naive way of storing the conditional probabilities of 10 two-valued variables as a table requires storage space for $2^{10} = 1024$

values. If no variable's local distribution depends on more than three parent variables, the Bayesian network representation stores at most $10 \cdot 2^3 = 80$ values. One advantage of Bayesian networks is that it is intuitively easier for a human to understand (a sparse set of) direct dependencies and local distributions than complete joint distributions.

Section 3

Local Markov property X is a Bayesian network with respect to G if it satisfies the local Markov property: each variable is conditionally independent of its non-descendants given its parent variables:

Section 4

$$\Pr(R, G | \text{do}(S = T)) = \Pr(R) \Pr(G | R, S = T) \\ \Pr(\text{do}\{S=T\}) = \Pr(R) \Pr(G | R, S=T)$$

Section 5

To answer an interventional question, such as "What is the probability that it would rain, given that we wet the grass?" the answer is governed by the post-intervention joint distribution function

Section 6

OpenBUGS – Open-source development of WinBUGS. SPSS Modeler – Commercial software that includes an implementation for Bayesian networks. Stan (software) – Stan is an open-source package for obtaining Bayesian inference using the No-U-Turn sampler (NUTS), a variant of Hamiltonian Monte Carlo. WinBUGS – One of the first computational implementations of MCMC samplers. No longer maintained.

Section 7

$$\Pr(R = T | G = T) = \Pr(G = T, R = T) \Pr(G = T) = \sum_{x \in \{T, F\}} \Pr(G = T, S = x, R = T) \sum_{x, y \in \{T, F\}} \Pr(G = T, S = x, R = y) \\ \Pr(R = T | G = T) = \frac{\Pr(G = T, R = T)}{\Pr(G = T)} = \frac{\sum_{x \in \{T, F\}} \Pr(G = T, S = x, R = T)}{\sum_{x, y \in \{T, F\}} \Pr(G = T, S = x, R = y)}$$

Section 8

$$\Pr(Y, Z | \text{do}(X)) = \Pr(Y, Z, X = x) \Pr(X = x | Z) \\ \Pr(Y, Z | \text{do}(X)) = \Pr(Y, Z, X = x) \Pr(X = x | Z)$$

Section 9

$$\Pr(G = T, S = T, R = T) = \Pr(G = T | S = T, R = T) \Pr(S = T | R = T) \Pr(R = T) = 0.99 \times 0.01 \times 0.2 = 0.00198 \\ \Pr(G = T, S = T, R = T) = \Pr(G = T | S = T, R = T) \Pr(S = T | R = T) \Pr(R = T) = 0.99 \times 0.01 \times 0.2 = 0.00198$$

$0.01 \times 0.2 \times 0.00198$

Section 10

$$\Pr(G, S, R) = \Pr(G \mid S, R) \Pr(S \mid R) \Pr(R)$$

Section 11

Factorization definition X is a Bayesian network with respect to G if its joint probability density function (with respect to a product measure) can be written as a product of the individual density functions, conditional on their parent variables:

Section 12

Suppose we want to model the dependencies between three variables: the sprinkler (or more appropriately, its state - whether it is on or not), the presence or absence of rain and whether the grass is wet or not. Observe that two events can cause the grass to become wet: an active sprinkler or rain. Rain has a direct effect on the use of the sprinkler (namely that when it rains, the sprinkler usually is not active). This situation can be modeled with a Bayesian network (shown to the right). Each variable has two possible values, T (for true) and F (for false). The joint probability function is, by the chain rule of probability,

Section 13

P contains (but does not need to be entirely) a directed chain, $u \leftarrow m \leftarrow v$ or $u \rightarrow m \rightarrow v$, such that the middle node m is in Z, P contains a fork, $u \leftarrow m \rightarrow v$, such that the middle node m is in Z, or P contains an inverted fork (or collider), $u \rightarrow m \leftarrow v$, such that the middle node m is not in Z and no descendant of m is in Z. The nodes u and v are d-separated by Z if all trails between them are d-separated. If u and v are not d-separated, they are d-connected. X is a Bayesian network with respect to G if, for any two nodes u, v: